

Auditing gender Bias in New SLM Models for IEEE Journals

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Abstract—The abstract goes here.

Index Terms—IEEE, IEEEtran, journal, LATEX, paper, template.

I. ABSTRACT

THE Small language models (SLMs) are increasingly used in real-world applications but can encode and amplify gender bias due to compact architectures and constrained training data. We present an end-to-end, reproducible auditing framework for detecting and quantifying gender bias in newly released SLMs, implemented in the EquiLens platform. Our approach combines template-driven and data-driven corpus generation that targets occupational and attribute associations, controlled model evaluations with configurable decoding and safety parameters, and a rigorous analysis pipeline that produces bias scores, statistical significance tests, and visualizations. EquiLens integrates with local model-serving infrastructure (Ollama), supports GPU-accelerated inference for scalable experiments, and provides robust interruption-and-resume semantics and a dual-auditor design (production and experimental) to balance reliability and exploratory metrics. Using this platform, we evaluate multiple SLM variants and show consistent gender-linked disparities across prompts and templates, with effect sizes that depend on model architecture and decoding settings. We discuss actionable mitigation strategies—prompt engineering, targeted fine-tuning, and post-hoc corrections—and highlight the operational trade-offs between audit sensitivity, runtime cost, and measurement uncertainty. We argue that routine, automated bias audits with reproducible toolchains like EquiLens should be standard practice for responsibly deploying SLMs in sensitive contexts.

August 26, 2025

A. Introduction

Small language models (SLMs) are increasingly deployed across real-world applications because they offer efficient inference and lower infrastructure costs. However, their compact architectures and often narrower training data make them especially vulnerable to encoding and amplifying social biases, including gender stereotyping. This paper addresses the urgent need for rigorous, reproducible auditing of gender bias in newly released SLMs by presenting an end-to-end evaluation

framework and empirical results that highlight common failure modes and actionable mitigation directions..

We adopt a systematic auditing pipeline that (1) generates targeted evaluation corpora using template-driven and data-driven methods that explicitly probe gendered associations with occupational and attribute-target words, (2) executes controlled model evaluations against these corpora under configurable sampling and safety margins, and (3) analyzes outcomes with quantitative bias scores, statistical significance tests, and visualizations to surface distributional and contextual bias patterns.

The pipeline is implemented as a modular platform designed for reproducibility: configuration files record model settings (temperature, max tokens), audit parameters (batch size, checkpoint frequency), and bias templates (example gender pronoun sets and target occupations), while each audit run produces a session directory containing raw responses, timing logs, summary statistics, and bias-visualization artifacts. Key engineering features that enable large-scale, reliable auditing include tight integration with local model-serving infrastructure, GPU-accelerated inference for throughput, robust interruption-and-resume semantics for long-running experiments, and a dual-auditor design that separates a production-hardened auditor from an experimental enhanced auditor for exploratory metrics. These capabilities permit fair comparisons across SLM variants and configurations (e.g., model size, decoding settings) and enable transparent reporting of runtime cost and measurement uncertainty—critical for reproducibility and operational decision-making. Empirically, our results demonstrate measurable gender-linked disparities across multiple SLMs and prompt templates, with effect sizes that vary by model architecture and decoding parameters. We discuss how these findings inform model selection, prompt engineering, dataset curation, and the design of mitigation strategies (e.g., targeted fine-tuning, post-processing bias corrections).

Finally, we emphasize broader implications for responsible deployment: routine, automated bias audits should be an integral part of the life cycle for SLMs used in sensitive contexts, and audit tool chains must prioritize reproducibility, interpretability, and actionable recommendations for model developers and policymakers.

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II. CONCLUSION

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APPENDIX A

PROOF OF THE FIRST ZONKLAR EQUATION

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APPENDIX B

Appendix two text goes here.

ACKNOWLEDGMENT

The authors would like to thank...

REFERENCES

- [1] H. Kopka and P. W. Daly, *A Guide to L^AT_EX*, 3rd ed. Harlow, England: Addison-Wesley, 1999.

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Jane Doe Biography text here.